Keep the conclusion within the evidence

Notice when a conclusion reaches beyond the people or results actually studied.

Suggested teaching time: 20 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None; learners compare a stated group with the group named in a conclusion.

Students will be able to

- Identify a change from the group studied to a wider group.
- Rewrite an overclaim without losing the result actually reported.
- Name evidence that would address the missing step.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; the instructor guide stays separate from student materials.
- Read all three scenarios and their model answers.
- Print the handout or display one activity at a time. Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

3 min	**Compare the two groups.** Learners complete the club warm-up. Ask them to underline who was asked and who the leader makes a claim about.
7 min	**Repair the school-wide claim.** Pairs complete the grading-rubric task. Check that rewritten conclusions include both the art class and the reported split in answers.
5 min	**Discuss the missing step.** Reveal the main answer and use the discussion prompts. Distinguish an estimate about a wider group from a claim that every member agrees.
5 min	**Transfer the skill.** Learners complete the residents survey independently. Use the assessment criteria before showing the model answer.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / A club speaks for everyone?

Student task and instructor answer

Explain these words as needed

Conclusion: The statement an argument asks us to accept on the basis of its reasons.

Sample: The people or cases actually studied.

Representative sample: A group selected to reflect the wider population relevant to the question.

Scenario

Invented scenario: Nine of ten chess-club members asked say they want a lunchtime tournament. The club leader says, 'Every student in our school wants one.' No students outside the club were asked.

Student output: Write one sentence identifying the jump and one narrower conclusion.

1. Name the group asked and the group covered by the leader's conclusion.

2. Write a conclusion that keeps both the surveyed group and the nine-of-ten result.

Student success check: Does your rewrite avoid both “all club members” and “all students”?

Model answer

The survey covers ten chess-club members, but the conclusion covers every student. Nine of the ten club members asked wanted a lunchtime tournament.

Why this meets the task: It preserves the observed result without extending it to unasked students or treating nine as ten.

Main activity / The grading-rubric survey

Student task and instructor answer

Scenario

Invented scenario: Eighteen of 24 students in one art class say they prefer a new grading rubric. The other six say they do not prefer it. The principal announces, 'Every student in our 600-student school prefers it.' No other students were asked.

Student output: Write a scope comparison, a supported conclusion, and one further evidence need.

1. Complete: 'The evidence covers ___, but the conclusion covers ___.'

2. Rewrite the conclusion as one sentence that the survey actually supports.

3. Name one additional kind of evidence needed before estimating school-wide preferences.

Student success check: Does your rewrite retain the art-class sample and the six students who did not prefer the rubric?

Model answer

The evidence covers 24 students in one art class; the conclusion covers all 600 students. A majority of the surveyed art class, 18 of 24 students, preferred the new rubric. Estimating school-wide preferences would need a representative sample across classes. Even that would not establish that every student agrees.

Why this meets the task: It narrows the conclusion to the evidence and distinguishes an estimate of wider preferences from a claim about every person.

Exit ticket / A survey of park volunteers

Student task and instructor answer

Scenario

Invented scenario: A town has 2,000 residents. Twelve of 15 residents at a park clean-up say they favor Sunday street closures. The mayor says, 'Most town residents favor the closures.' No other residents were asked.

Student output: Write three sentences: the gap, a supported conclusion, and a better evidence plan.

1. Compare the residents actually asked with those covered by the mayor's conclusion.

2. Rewrite the result without extending it beyond those surveyed.

3. Describe who a new survey should include to estimate town-wide support.

Student success check: Have you noticed that even a claim about “most” can reach beyond an unrepresentative sample?

Model answer

The survey covers 15 clean-up participants, not the town as a whole. Twelve of those 15 participants favored Sunday closures. A town-wide estimate should use a representative sample of residents, including people who do not volunteer at park clean-ups.

Why this meets the task: It does not assume volunteers reflect all residents, yet retains the information their answers actually supply.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The sample and wider claimed group are identified accurately.
- The rewrite preserves the number, group, and limits of the reported result.
- The proposed evidence addresses the wider population without promising certainty about everyone.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Does a narrow sample alone prove a wider conclusion false?

Listen for: No. A sample can fail to establish a conclusion without refuting it. In this art-class case, however, six students explicitly disagree, so the claim that every student prefers it is false.

Why would a larger sample from the same art class still leave a problem?

Listen for: Including more people does not by itself fix a mismatch between the group studied and the wider population.

Common confusions

A majority in a sample means a majority everywhere. Ask how the sample was chosen and which population it can reasonably represent.

A gap makes the whole survey worthless. The survey still describes the people asked. Limit the conclusion to what it supports.

Representative sampling proves everyone agrees. It can support an estimate of a population pattern, with uncertainty, not a guarantee about every person.

Adapt the lesson

More support: Use two boxes: 'Who was asked?' and 'Who is the claim about?' For the main task, expect '24 art students' and 'all 600 students'. Then retain '18 of 24' in the rewrite.

More challenge: Ask whether the art-class result refutes 'every student prefers it'. Expected answer: yes, the six surveyed students who did not prefer it already contradict that universal claim. This goes beyond merely noting that the sample is narrow.

Next practice

https://xixteen.com/learners/?practice=logical_gaps#workbench